Equivalence of Rényi DP, (ε, δ)-DP, and f-DP Curves

1 Setup and Assumptions

Consider the additive-noise mechanism $M(x) = A(x) + Z$ with $Z \sim \nu$ centered and symmetric. For neighboring datasets x, x', let

$$P = \mathcal{L}(M(x)), \qquad Q = \mathcal{L}(M(x')).$$

We work under the following regularity assumptions, which hold for essentially all noise distributions of practical interest (Gaussian, Laplace, etc.):

(A1) P and Q are mutually absolutely continuous with densities p, q on $\mathbb{R}^d$.

(A2) The privacy loss random variable

$$L = \log \frac{p(Y)}{q(Y)}, \qquad Y \sim P,$$

has a distribution with no atoms, and the moment generating function $\mathbb{E}_Q[(p/q)^\alpha]$ is finite for all $\alpha > 1$ in a neighborhood of every α where we evaluate.

(A3) The three "DP curves" are:

- **Rényi:** $R(\alpha) := D_\alpha(P\|Q) = \frac{1}{\alpha-1} \log \mathbb{E}_Q[(p/q)^\alpha]$, $\alpha > 1$.
- (ε, δ)**-DP:** $\delta(\varepsilon) := \sup_S \big(P(S) - e^\varepsilon Q(S)\big) = \mathbb{E}_Q[(p/q - e^\varepsilon)_+]$, $\varepsilon > 0$.
- f**-DP:** the trade-off function $T(P, Q)(\alpha) = \inf\{1 - \mathbb{E}_Q[\phi] : \mathbb{E}_P[\phi] \geq \alpha,\ 0 \leq \phi \leq 1\}$, $\alpha \in [0, 1]$.

For symmetry, the *symmetric* versions are obtained by also considering $D_\alpha(Q\|P)$, $\delta(\varepsilon)$ computed from the worst of (P, Q) and (Q, P), and $f(\alpha) = \min\{T(P, Q), T(Q, P)\}$. Since ν is symmetric and A is deterministic, one can check that P and Q are exchangeable up to translation, so the curves for (P, Q) and (Q, P) coincide; we work with the one-sided versions below.

2 Main Result

Theorem 1. *Under (A1)–(A3), the following three pieces of information are equivalent:*

1. *The full Rényi DP curve* $\{R(\alpha)\}_{\alpha>1}$.
2. *The full* (ε, δ)*-DP curve* $\{\delta(\varepsilon)\}_{\varepsilon>0}$.
3. *The full* f*-DP curve* $\{f(\alpha)\}_{\alpha\in[0,1]}$.

More precisely, each curve determines the distribution of the privacy loss random variable L under Q, and hence determines the other two curves.

The proof has two parts: first we show (2) $\Leftrightarrow$ (3) via classical convex-duality arguments (this is well known, but we include it for completeness), and then we establish the new content: (1) $\Leftrightarrow$ (2).

3 The Privacy Loss Distribution as a Common Sufficient Statistic

Let $L = \log(p/q)$ viewed as a random variable, and let

$$\mu(t) := Q(L \le t), \qquad \tilde{\mu}(t) := P(L \le t)$$

be its CDFs under Q and P respectively. A standard change of measure gives

$$\tilde{\mu}(\mathrm{d}t) = e^t \, \mu(\mathrm{d}t), \tag{$\star$}$$

because $\frac{\mathrm{d}P}{\mathrm{d}Q} = e^L$ Q-almost surely. Consequently μ alone determines $\tilde{\mu}$.

Lemma 1. *Each of the three DP curves is a functional of μ alone.*

Proof. Using ($\star$):

- $\mathbb{E}_Q[(p/q)^\alpha] = \int e^{\alpha t} \, \mu(\mathrm{d}t)$, so $R(\alpha) = \frac{1}{\alpha - 1} \log \int e^{\alpha t} \mu(\mathrm{d}t)$.
- $\delta(\varepsilon) = \mathbb{E}_Q[(e^L - e^\varepsilon)_+] = \int (e^t - e^\varepsilon)_+ \, \mu(\mathrm{d}t)$.
- By the Neyman–Pearson lemma, the trade-off function f is determined by the distribution of L under P and Q, hence by μ via ($\star$). □

Thus to prove the theorem it suffices to show that **each curve uniquely determines μ.**

4 (ε, δ)-DP Curve $\Leftrightarrow$ Privacy Loss Distribution

Proposition 1. *Under (A1)–(A2), the function $\varepsilon \mapsto \delta(\varepsilon)$ uniquely determines μ.*

Proof. From ($\star$),

$$\delta(\varepsilon) = \int_\varepsilon^\infty (e^t - e^\varepsilon) \, \mu(\mathrm{d}t) = \int_\varepsilon^\infty e^t \, \mu(\mathrm{d}t) - e^\varepsilon \, \mu([\varepsilon, \infty)).$$

Differentiating in ε (legitimate since (A2) ensures the integrand is absolutely continuous and μ has no atoms), we obtain

$$\frac{\mathrm{d}\delta}{\mathrm{d}\varepsilon}(\varepsilon) = -e^\varepsilon \, \mu([\varepsilon, \infty)) \quad \Longleftrightarrow \quad \mu([\varepsilon, \infty)) = -e^{-\varepsilon} \, \delta'(\varepsilon).$$

For $\varepsilon < 0$ (extending the curve in the natural way), the same computation yields μ on $(-\infty, 0)$ as well; alternatively, ($\star$) implies $\mu((-\infty, -\varepsilon]) = e^{-\varepsilon} \tilde{\mu}((-\infty, -\varepsilon])$, which by the symmetry of the problem can be read off from the curve for (Q, P). Either way, μ is fully recovered from $\delta(\cdot)$.

Conversely, μ determines δ by Lemma 1. □

The equivalence (2) $\Leftrightarrow$ (3) is the well-known Dong–Roth–Su correspondence: the trade-off function f is the Legendre-type dual of the (ε, δ) curve,

$$f(\alpha) = \sup_{\varepsilon \ge 0} \big(1 - \delta(\varepsilon) - e^\varepsilon \alpha\big),$$

and this transform is an involution on convex DP curves. We omit the details.

5 Rényi DP Curve $\Leftrightarrow$ Privacy Loss Distribution

This is the substantive content of the theorem.

Proposition 2. *Under (A1)–(A2), the function $\alpha \mapsto R(\alpha)$ on $(1,\infty)$ uniquely determines μ.*

Proof. Define the moment generating function of μ:

$$\Phi(\alpha) := \int_{\mathbb{R}} e^{\alpha t}\, \mu(\mathrm{d}t), \qquad \alpha > 1.$$

By Lemma 1, $R(\alpha) = \frac{1}{\alpha-1}\log\Phi(\alpha)$, so $\Phi(\alpha) = \exp\big((\alpha-1)R(\alpha)\big)$ is determined on $(1,\infty)$.

Step 1: Extension to a strip. Assumption (A2) and Hölder's inequality imply Φ is finite on an open interval containing $(1,\infty)$. Moreover, Φ extends to a holomorphic function on the strip $\{z \in \mathbb{C} : 1 < \mathrm{Re}(z) < \alpha_{\max}\}$ for any $\alpha_{\max}$ in the interior of the domain of finiteness, because $\partial_z e^{zt} = te^{zt}$ is bounded by $|t|e^{(\mathrm{Re}\, z)t}$, which is μ-integrable on compact substrips.

Step 2: Identification by analytic continuation. Suppose μ_1, μ_2 are two probability measures on $\mathbb{R}$ both consistent with the same Rényi curve R. Their MGFs Φ_1, Φ_2 agree on $(1,\infty)$, hence by the identity theorem for holomorphic functions, they agree on the entire common strip of analyticity.

Step 3: Uniqueness of μ from MGF on a strip. Pick any $\alpha_0 > 1$ in the interior of this strip. The function $h \mapsto \Phi_j(\alpha_0 + ih)$ is the Fourier transform of the finite measure $e^{\alpha_0 t}\mu_j(\mathrm{d}t)$. Since $\Phi_1 \equiv \Phi_2$ on the vertical line $\mathrm{Re}(z) = \alpha_0$, the Fourier transforms of these two finite measures coincide. By Fourier uniqueness,

$$e^{\alpha_0 t}\mu_1(\mathrm{d}t) = e^{\alpha_0 t}\mu_2(\mathrm{d}t),$$

and dividing by the strictly positive factor $e^{\alpha_0 t}$ gives $\mu_1 = \mu_2$. □

Corollary 1. *Combining Propositions 1 and 2 with the $(\varepsilon,\delta) \Leftrightarrow f$-DP correspondence, all three DP curves carry exactly the same information, namely the law of the privacy loss random variable under Q. Hence each one determines the other two.*

6 Remarks on Regularity and Failure Modes

Both directions of the equivalence rely crucially on (A2). The following two points clarify the scope.

(i) Need for the full range $\alpha > 1$. Knowing $R(\alpha)$ only on a finite interval $(1, \alpha_0)$ is *not* sufficient in general: a bounded set of moments does not determine a distribution with unbounded support. The proof of Proposition 2 uses that the Rényi curve is known on an interval containing a *limit point inside its analyticity domain*, which yields the MGF on a full strip via analytic continuation.

(ii) Why $(\star)$ suffices for both directions. The key structural fact is that the privacy loss random variable's distribution under Q is a complete sufficient statistic for the dominated pair (P, Q) as far as DP-style hypothesis testing is concerned. This is the underlying reason — visible in the trade-off function literature — why all three curves are equivalent.

(iii) Symmetric additive mechanisms. For the mechanism $M(x) = A(x) + Z$ with Z symmetric, P and Q differ by a translation $\Delta = A(x') - A(x)$, so $L = \log\frac{\nu(y)}{\nu(y-\Delta)}$ under $Y \sim P$. Symmetry of ν implies μ and $\tilde{\mu}$ are related by $\tilde{\mu}(t) = 1 - \mu(-t)$, recovering the standard symmetry of the f-DP curve $f(\alpha) = f^{-1}(\alpha)$.

7 Conclusion

Answer to the problem. *Yes, and yes.* Under standard regularity (A1)–(A3) — which is satisfied by all common additive-noise mechanisms such as Gaussian and Laplace — the full exact Rényi DP curve, the full exact (ε, δ)-DP curve, and the full exact f-DP curve are pairwise equivalent: each one uniquely determines the privacy loss random variable's distribution and hence the other two curves. The implications hold in **both** directions; there is no informational asymmetry among the three formalisms when the curves are known exactly and in full.

The conversion maps are explicit:

$$\mu([\varepsilon, \infty)) = -e^{-\varepsilon}\delta'(\varepsilon), \qquad \Phi(\alpha) = e^{(\alpha-1)R(\alpha)} = \int e^{\alpha t}\mu(\mathrm{d}t),$$

with μ then yielding f via the Neyman–Pearson lemma applied to the likelihood ratio e^L.